

Power Series Solutions to Nonlinear Ordinary Differential Equations

MATH 689 Homework 5

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1. Apply the identity for raising a series to a power to show that the Taylor expansion of

$$f = (1 + x^2 + e^x)^{1/2} \quad (1)$$

is given as

$$\sum_{n=0}^{\infty} b_n x^n \quad (2)$$

where

$$b_{n>0} = \frac{1}{2n} \sum_{j=1}^n \left(\frac{3}{2}j - n \right) a_j b_{n-j}, \quad b_0 = \sqrt{2} \quad (3)$$

with

$$a_j = \begin{cases} 1, & j = 1 \\ 3/2, & j = 2 \\ 1/j!, & j \geq 3 \end{cases} \quad (4)$$

Solution.

Recall the identity for a power series raised to the power γ ,

$$b_{n>0} = \frac{1}{n a_0} \sum_{j=1}^n (j\gamma - n + j) a_j b_{n-j}, \quad b_0 = (a_0)^\gamma \quad (5)$$

For our problem $f = (1 + x^2 + e^x)^{1/2}$, so $\gamma = 1/2$.

We write $1 + x^2 + e^x$ as $\sum_{n=0}^{\infty} a_n x^n$. Expanding e^x and collecting terms gives

$$1 + x^2 + e^x = 1 + x^2 + \sum_{0 \leq n \leq \infty} \frac{x^n}{n!} = (1 + 1) + x + \left(x^2 + \frac{x^2}{2}\right) + \sum_{3 \leq n \leq \infty} \frac{x^n}{n!} \quad (6)$$

so the coefficients are

$$a_0 = 2, \quad a_1 = 1, \quad a_2 = 1 + \frac{1}{2} = \frac{3}{2}, \quad a_j = \frac{1}{j!} \text{ for } j \geq 3. \quad (7)$$

Then

$$b_0 = (a_0)^\gamma = 2^{1/2} = \sqrt{2}. \quad (8)$$

Substituting $\gamma = 1/2$ and $a_0 = 2$

$$b_{n>0} = \frac{1}{n \cdot 2} \sum_{j=1}^n \left(\frac{j}{2} - n + j \right) a_j b_{n-j} = \frac{1}{2n} \sum_{j=1}^n \left(\frac{3}{2}j - n \right) a_j b_{n-j} \quad (9)$$

For $n = 1$:

$$b_1 = \frac{1}{2(1)} \left(\frac{3}{2}(1) - 1 \right) a_1 b_0 = \frac{1}{2} \cdot \frac{1}{2} \cdot 1 \cdot \sqrt{2} = \frac{\sqrt{2}}{4}. \quad (10)$$

For $n = 2$:

$$b_2 = \frac{1}{2(2)} \left[\left(\frac{3}{2}(1) - 2 \right) a_1 b_1 + \left(\frac{3}{2}(2) - 2 \right) a_2 b_0 \right] \quad (11)$$

$$= \frac{1}{4} \left[\left(-\frac{1}{2} \right) (1) \frac{\sqrt{2}}{4} + (1) \left(\frac{3}{2} \right) \sqrt{2} \right] = \frac{1}{4} \left[-\frac{\sqrt{2}}{8} + \frac{3\sqrt{2}}{2} \right] = \frac{11\sqrt{2}}{32}. \quad (12)$$

We can confirm this coefficient from expanding the binomial coefficients in the equation of f .

2. Analytic Continuation of the Flierl-Petviashvili (FP) Problem

- (a) Look back at Problem 2 of HW3. Ensure you have a working code such that the series solution matches the numerical solution in a region near $r = 0$, prior to the radius of convergence.
- (b) Save this as a new file for HW5 (and to preserve your old code for HW3). Comment or delete the code chunk beginning with `figure (2)` that you added before. Everything will now be done in one figure window (`figure 1`).
- (c) Run your code and note that the numerical solution of the FP problem has the behavior $f \rightarrow 0$ as $x \rightarrow \infty$. The original problem is stated as a BVP:

$$f'' + \frac{1}{x}f' - f - f^2 = 0, \quad f'(0) = 0, \quad f(\infty) = 0. \quad (13)$$

The power series found in HW3,

$$f = \sum_{n=0}^{\infty} a_n x^n, \quad a_{n+2} = \frac{a_n + \sum_{j=0}^n a_j a_{n-j}}{(n+2)^2}, \quad a_1 = 0, \quad a_0 = -2.392, \quad (14)$$

has a radius of convergence $|x| < 2.6$ (approximately). Construct a resummation of the original series as follows

$$f = \sum_{n=0}^{\infty} a_n x^n = \frac{1}{\sum_{n=0}^{\infty} A_n x^n}, \quad (15)$$

where the A_n coefficients are obtained in terms of the known a_n coefficients.

Solution. We derive the resummation for $F(\cdot)$ using the Cauchy product

$$\begin{aligned} \sum_{0 \leq n \leq \infty} A_n x^n \left(\sum_{0 \leq n \leq \infty} a_n x^n \right) &= 1 \\ \sum_{0 \leq n \leq \infty} \left(\sum_{0 \leq j \leq n} A_{n-j} a_j \right) x^n &= 1 \end{aligned} \quad (16)$$

then we take out the A_n term by $j = 0$ and for $n > 0$ we have

$$\begin{aligned} A_n a_0 + \sum_{1 \leq j \leq n} A_{n-j} a_j &= 0 \\ A_n &= \frac{-\sum_{1 \leq j \leq n} A_{n-j} a_j}{a_0}, \quad A_0 = 1/a_0, \quad A_1 = -a_1/a_0^2 \end{aligned} \quad (17)$$

Notice that for A_2 we need $a_2 = (a_0 + a_0^2)/4$ which was not solved for explicitly before.

- (d) Incorporate the above resummation into your code, somewhere you have computed the original series coefficients.

MATLAB - Chuck 1

```
figure(1)
dx=0.0001; %STEP SIZE
x=dx:dx:10; %DOMAIN OF INDEPENDENT VARIABLE
f=x;
u0=[-2.392, 0];
%NUMERICAL SOLUTION (RK4 ENGINE)
f(1)=u0(1); u=u0;
for n=1:length(x)-1
    k1=F(u,x(n)) ; k2=F(u+k1*dx/2,x(n)+dx/2) ;
    k3=F(u+k2*dx/2,x(n)+dx/2) ; k4=F(u+k3*dx,x(n)+dx) ;
    u=u+(dx/6)*(k1+2*k2+2*k3+k4); f(n+1)=u(1);
end; hold on

%PLOT NUMERICAL SOLUTION, DOWNSAMPLE POINTS
plot(x(1:1e3:end),f(1:1e3:end),'k.','markersize',10,'displayname','Numerical (RK4)',
, Color= "m");
hold on
legend show; xlabel('$x$', 'Interpreter','latex','FontSize',10)
ylabel('$f$', 'Interpreter','latex','FontSize',20)
% 1st order ODE SYSTEM (or single equation, if 1st order already)
function dudx = F(u,x)
f=u(1); v=u(2);
dudx=[v f^2+f-v/x];
end
%coefficient(s) that come(s) from initial condition(s)
N=10;
a0 = -2.392; % a0
a(1) = 0; % a1
a(2) = (a0+a0^2)/4;
for n = 1:N-1
    asum = a0*a(n) + a(n)*a0 ;% j=0 and j=n terms
    for j = 1:n-1
        asum = asum + a(j)*a(n-j);
    end
    a(n+2) = (a(n) + asum)./(n+2).^2;
end
%PLOT THE POWER SERIES SOLUTION
fS=a0; axis1=axis;
for n=1:N
    fS=fS+a(n)*x.^(n);
    if round(n/2)== n/2
        plot(x,fS,'--','LineWidth',2,'DisplayName',[num2str(n) ' term series']);
    end
end; axis(axis1)
legend('location','northeast','fontsize',10)
```

```

%RESUMMATION COEFFICIENTS
A0 = 1/a0;
A(1) = -a(1)/a0^2;

for n=2:N
    Asum = A0*a(n);
    for j=1:n-1
        Asum = Asum + A(j)*a(n-j);
    end
    A(n) = -Asum/a0;
end

%PLOT THE RESUMMATION
ax=gca; ax.ColorOrderIndex=1;
fS=a0; axis1=axis;
for n=1:N
    fS=fS+a(n)*x.^n;
    if round(n/2)==n/2
        plot(x,fS,'--','LineWidth',2,'DisplayName',[num2str(n) ' term series'])
    end
end; axis(axis1)

%PLOT THE RESUMMATION
ax=gca; ax.ColorOrderIndex=1;
fAsum=A0;
for n=1:N
    fAsum=fAsum+A(n)*x.^n;
    fA=1./fAsum;
    if round(n/2)==n/2
        plot(x,fA,'LineWidth',2,'DisplayName',[num2str(n) ' term resum']);
    end
end; axis(axis1)
legend('location','southeast' ); legend show

```

(e) Plot truncations of the resummation on the same plot as the series and numerical solutions for $N = 2, 4, 6, 8, 10$.

Solution / Plot.

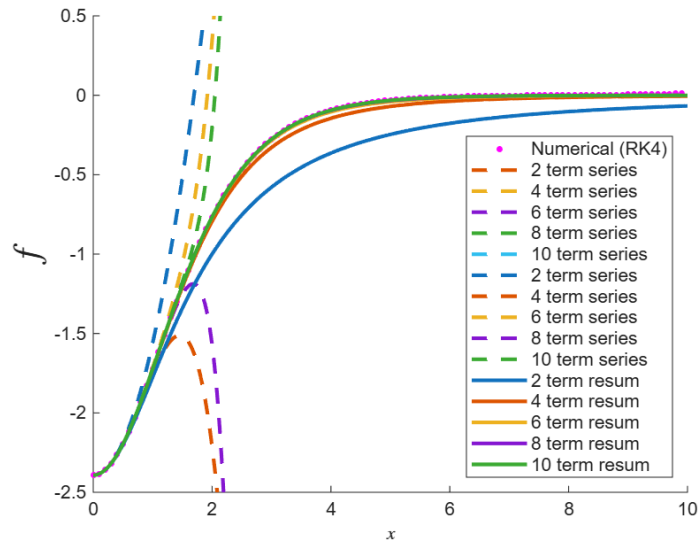


Figure 1: Resummation of the FP Problem

3. (MATH689 Problem)

- (a) Noting that $\cosh(f) \equiv (e^f + e^{-f})/2$ and $\sinh(f) \equiv (e^f - e^{-f})/2$, develop a way of recursively computing the hyperbolic cosine of a series. That is, if

$$\cosh\left(\sum_{n=0}^{\infty} a_n x^n\right) = \sum_{n=0}^{\infty} c_n x^n, \quad (18)$$

find the c_n coefficients given the a_n coefficients.

Solution.

Using the identity, we have

$$\begin{aligned} \cosh(f) &= \frac{1}{2} \left(\exp\left(\sum_{0 \leq n \leq \infty} a_n x^n\right) + \exp\left(-\sum_{0 \leq n \leq \infty} a_n x^n\right) \right) \\ &= \frac{1}{2} \left(\exp\left(\sum_{0 \leq n \leq \infty} a_n x^n\right) + \left[\exp\left(\sum_{0 \leq n \leq \infty} a_n x^n\right) \right]^{-1} \right) \\ &= \frac{1}{2} \left(\sum_{0 \leq n \leq \infty} g_n x^n + \sum_{0 \leq n \leq \infty} t_n x^n \right) \\ &= \sum_{0 \leq n \leq \infty} c_n x^n \end{aligned} \quad (19)$$

Then write

$$\sum_{0 \leq n \leq \infty} g_n x^n = \exp\left(\sum_{0 \leq n \leq \infty} a_n x^n\right), \quad \left[\sum_{0 \leq n \leq \infty} g_n x^n \right]^{-1} = \sum_{0 \leq n \leq \infty} t_n x^n \quad (20)$$

To derive this we take the first derivative

$$\begin{aligned} \sum_{0 \leq n \leq \infty} g_{n+1}(n+1)x^n &= \sum_{0 \leq n \leq \infty} a_{n+1}(n+1)x^n \left[\exp\left(\sum_{0 \leq n \leq \infty} a_n x^n\right) \right] \\ \sum_{0 \leq n \leq \infty} g_{n+1}(n+1)x^n &= \sum_{0 \leq n \leq \infty} a_{n+1}(n+1)x^n \left[\sum_{0 \leq n \leq \infty} g_n x^n \right] \end{aligned} \quad (21)$$

Via the Cauchy product again

$$g_{n+1} = \frac{1}{n+1} \sum_{0 \leq j \leq n} a_{j+1}(j+1)g_{n-j}, \quad g_0 = e^{a_0} \quad (22)$$

Then using the power identity for $\gamma = -1$ derived above in *problem 1* yields

$$t_{n>0} = \frac{-1}{g_0} \sum_{1 \leq j \leq n} g_j t_{n-j}, \quad t_0 = 1/g_0, \quad t_1 = -g_1/g_0^2 \quad (23)$$

(b) Test out your formula(s) in (a) by computing c_0 through c_3 in the series

$$\cosh\left(\frac{1}{1+x}\right) = \sum_{n=0}^{\infty} c_n x^n, \quad (24)$$

noting that

$$\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n, \quad |x| < 1. \quad (25)$$

Solution. We have $f(x) = \frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n$, so

$$a_0 = 1, \quad a_1 = -1, \quad a_2 = 1, \quad a_3 = -1, \quad \dots$$

Recall the recurrences from part (a):

$$\begin{aligned} g_{n+1} &= \frac{1}{n+1} \sum_{j=0}^n (j+1) a_{j+1} g_{n-j}, \quad g_0 = e^{a_0} = e \\ t_n &= -\frac{1}{g_0} \sum_{j=1}^n g_j t_{n-j}, \quad t_0 = \frac{1}{g_0} = e^{-1} \\ c_n &= \frac{g_n + t_n}{2}. \end{aligned}$$

Using these, we compute:

$$\begin{aligned} g_0 &= e, \\ g_1 &= (1)a_1 g_0 = (-1)e = -e, \\ g_2 &= \frac{1}{2} [(1)a_1 g_1 + (2)a_2 g_0] = \frac{1}{2} [(-1)(-e) + (2)(1)e] = \frac{1}{2}(e + 2e) = \frac{3e}{2}, \\ g_3 &= \frac{1}{3} [(1)a_1 g_2 + (2)a_2 g_1 + (3)a_3 g_0] \\ &= \frac{1}{3} \left[(-1)\frac{3e}{2} + (2)(1)(-e) + (3)(-1)e \right] = \frac{1}{3} \left(-\frac{3e}{2} - 2e - 3e \right) = -\frac{13e}{6}. \end{aligned}$$

Now compute t_n :

$$\begin{aligned} t_0 &= e^{-1}, \\ t_1 &= -\frac{1}{g_0} g_1 t_0 = -\frac{1}{e} (-e) \frac{1}{e} = \frac{1}{e}, \\ t_2 &= -\frac{1}{g_0} (g_1 t_1 + g_2 t_0) = -\frac{1}{e} \left((-e) \frac{1}{e} + \frac{3e}{2} \cdot \frac{1}{e} \right) = -\frac{1}{e} \left(-1 + \frac{3}{2} \right) = -\frac{1}{e} \cdot \frac{1}{2} = -\frac{1}{2e}, \\ t_3 &= -\frac{1}{g_0} (g_1 t_2 + g_2 t_1 + g_3 t_0) \\ &= -\frac{1}{e} \left((-e) \left(-\frac{1}{2e} \right) + \frac{3e}{2} \cdot \frac{1}{e} + \left(-\frac{13e}{6} \right) \cdot \frac{1}{e} \right) \\ &= -\frac{1}{e} \left(\frac{1}{2} + \frac{3}{2} - \frac{13}{6} \right) = -\frac{1}{e} \left(2 - \frac{13}{6} \right) = -\frac{1}{e} \left(\frac{12}{6} - \frac{13}{6} \right) = -\frac{1}{e} \left(-\frac{1}{6} \right) = \frac{1}{6e}. \end{aligned}$$

Finally, compute $c_n = (g_n + t_n)/2$:

$$\begin{aligned} c_0 &= \frac{e + e^{-1}}{2} = \cosh(1), \\ c_1 &= \frac{-e + e^{-1}}{2} = -\frac{e - e^{-1}}{2} = -\sinh(1), \\ c_2 &= \frac{\frac{3e}{2} - \frac{1}{2e}}{2} = \frac{3e}{4} - \frac{1}{4e} = \frac{3e^2 - 1}{4e} = \frac{\cosh(1)}{2} + \sinh(1), \\ c_3 &= \frac{-\frac{13e}{6} + \frac{1}{6e}}{2} = -\frac{13e}{12} + \frac{1}{12e} = -\frac{13e^2 - 1}{12e}. \end{aligned}$$

Simplifying c_3 further using $e^2 = \cosh(1) + \sinh(1)$ and $1/e = \cosh(1) - \sinh(1)$ yields $c_3 = -\frac{7}{6} \sinh(1) - \cosh(1)$.

Thus we obtain

$$\cosh\left(\frac{1}{1+x}\right) = \cosh(1) - x \sinh(1) + x^2 \left(\sinh(1) + \frac{\cosh(1)}{2} \right) + x^3 \left(-\frac{7 \sinh(1)}{6} - \cosh(1) \right) + O(x^4).$$

The series representation from wolfram alpha was given as

$$\cosh(1) - x \sinh(1) + x^2 \left(\sinh(1) + \frac{\cosh(1)}{2} \right) + x^3 \left(-\frac{7 \sinh(1)}{6} - \cosh(1) \right) + x^4 \left(\frac{3 \sinh(1)}{2} + \frac{37 \cosh(1)}{24} \right) + O(x^5),$$

where the x^4 term can be verified by extending our recurrence to $n = 4$ if needed.